

DBCLS BioHackathon 2026 report: StrideCheck, a browser-only prototype that turns running videos into gait measures, joint-load estimates and an honest injury prediction

Keitaro Takaoki¹, Haruma Abe¹, and Teppei Okazaki¹

¹ Tokyo City University, Tokyo, Japan 

BioHackathon series:

[DBCLS BioHackathon 2026](#)

Matsuyama, Japan, 2026

[StrideCheck](#)

Submitted: 18 Sep 2026

License:

Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC-BY](#)).

Published by [BioHackrXiv.org](#)

Introduction

Running-related injuries are common, and gait analysis that could flag risky running form usually needs a laboratory. Smartphone video is available to every runner, but two questions decide whether it is useful: how well can running form be measured from ordinary video, and how much do those measures actually tell about injury or joint load?

During the DBCLS BioHackathon 2026 we built StrideCheck, a prototype web page that answers both questions openly. From a side-view and a front- or rear-view video it (1) describes the runner's rhythm and form as types placed on the distribution of published cohorts, (2) gives a form-based injury prediction with its uncertainty and its (low) accuracy, and (3) estimates the load on four body regions relative to an average runner. The video is analysed entirely in the browser and never uploaded. Every model was built from open data, and the video measurement was validated against motion capture and force plates recorded with the same videos.

Data

We used four public datasets to build the models and two to validate the video measurement (Table 1).

Table 1: Datasets. The Loh datasets are CC BY-NC; the others are CC BY 4.0 or were used for validation only and are not redistributed.

Dataset	Content	Used for
Injury cohort (Wu et al., 2026)	142 endurance runners, 12-month prospective follow-up, weekly injury records, force-based gait timing at baseline	Rhythm types; rhythm-based prediction
Injury cohort (Loh et al., 2025)	81 runners (26 injured within 12 months), 2-D video angles at baseline	Form types; form-based prediction
Case-control (Loh & Kong, 2026)	154 injured and 44 uninjured runners, 2-D video angles	Angle definitions and distributions
Biomechanics (R. Fukuchi et al., 2023 ; R. K. Fukuchi et al., 2017)	39 runners, treadmill 2.5-4.5 m/s, markers and force plate	Joint-load models; validation of event detection

Dataset	Content	Used for
Validation (Wang, 2022; Wang et al., 2022, 2023)	Treadmill running at three speeds (6.3-9.9 km/h), front and side video (33 fps) recorded with Qualisys markers and a split-belt force treadmill	Validation of the video measurement (24 trials, 9 runners)

Methods

Types and injury prediction

To test whether types carry injury information, runners were clustered on cadence, duty factor, left-right asymmetry and foot strike (Wu et al.) and on form angles (Loh et al.). For display, the page uses simpler types: cadence by foot strike, and contralateral pelvic drop by knee flexion at mid-stance, each split at the weighted median of the cohort. The types only describe a runner; they are not used for prediction. Injury prediction uses the measured values directly (logistic models on pelvic drop, hip adduction, knee flexion and knee valgus), evaluated with person-wise cross-validation and bootstrap confidence intervals.

Joint load

From the Fukuchi markers we computed the values a video would give (markers projected to 2-D) and fitted ridge models for peak joint moments and ground reaction forces. For each region we report how much the video values explain beyond running speed and body size (R^2_{extra}) as a reliability rating.

Video measurement in the browser

The page runs MediaPipe Pose Landmarker Heavy (Bazarevsky et al., 2020) in the browser. The frame rate is read from the MP4 header, and every frame is analysed once. Initial contact is the heel's most forward point relative to the pelvis plus 29 ms, and toe-off is the toe's most rearward point minus 12% of the stride (Zeni et al., 2008); both offsets were fitted against force plates. In the side view the pose model often labels the legs by position (front and back) rather than by side, so the labels swap at every step and the far leg's points are pulled onto the near leg when the legs cross. StrideCheck tracks the two legs through each crossing (one crossing per step), predicts their path through the crossing frames, and names and corrects them step by step against the front or rear video, where left and right are reliable. A check view draws the estimated joints over the video so that users can see these steps.

Validation

Event detection was first checked on the Fukuchi markers, laid out as pose landmarks (hip joint centre as in (Harrington et al., 2007)), at 33-120 fps with added noise. The whole pipeline was then checked on the Wang et al. videos: the Qualisys recording starts with the videos (offset 3 ms, median), the force treadmill gives the true contacts of each leg, and the markers give the true angles, computed with the same 2-D formulas.

Results

Types and prediction. Neither rhythm types nor form types differed in injury rate (three rhythm clusters: 67%, 88% and 85% injured within 52 weeks, log-rank $p = 0.20$; form types: 28% vs 36%, $p = 0.44$). The form-based prediction reached an AUC of 0.64 [0.52-0.76]; the

rhythm-based one, 0.58 [0.39-0.75], cannot be told apart from chance. Adding injury history or training volume made prediction worse. The published AUC of 0.76 for the Wu et al. data was reproduced only when rows were split at random; split by person, it fell to 0.48.

Joint load. Beyond speed and body size, the video values explained part of the load for the shin (R^2_{extra} 0.37), the patellofemoral region (0.30) and the lateral hip (0.15), but not the Achilles tendon (-0.07). A 10% higher cadence corresponded to 8.7% lower knee load and 6.4% lower hip load.

Video measurement. Table 2 compares the browser pipeline with the ground truth. Leg tracking raised the share of steps with the correct left/right label from 91% (the pose model's own labels) to 98%. Initial contact was unbiased on both legs. Contact time, overstride and the near-leg foot angle at contact followed the truth closely. Knee flexion at mid-stance did not: runners differ in it by only 6.8° (SD), less than the measurement error, so the page does not report it and uses the cohort median instead.

Table 2: Browser pipeline (MediaPipe Heavy) against motion capture and force plates, 24 trials; the side camera filmed the left side, so the right leg is the far leg.

Measure (per trial and leg)	Near leg	Far leg
Initial contact per step	-1 ± 16 ms	+2 ± 12 ms
Contact time	bias -6 ms, SD 29 ms, $r = 0.83$	bias -4 ms, SD 20 ms, $r = 0.91$
Foot angle at contact	+5.3°, SD 4.1°, $r = 0.85$	+16.1°, SD 9.1°, $r = 0.61$
Overstride	$r = 0.93$	$r = 0.94$
Knee flexion at mid-stance	-16.6°, $r = 0.48$	-14.7°, $r = -0.01$

Discussion

StrideCheck shows how far ordinary video and open data go today. Timing measures and overstride can be measured from 33 fps video with errors that are small relative to differences between runners, whereas knee angles cannot. The injury predictions built on these measures are weak, and the page states their accuracy next to every number.

Limitations: the contact-time error (20-29 ms) is larger than the 10 ms target, which needs 120 fps video; the front video was used in place of a rear view; the far-leg foot angle is biased; one setting of the leg correction was chosen on the validation trials; and the injury cohorts are small. StrideCheck is a research prototype, not medical advice.

The next step is to turn the page into an entry point for prospective open data: runners who consent would share their measured values and later injuries, so that the models can be rebuilt on data that match the way the page measures.

Availability

The page runs at <https://keitaro0510.github.io/stride-check/>. Its source is at <https://github.com/Keitaro0510/stride-check>, and the analysis and validation code at <https://github.com/Keitaro0510/stride-check-analysis>, both under the MIT License. Third-party data are not redistributed; values derived from them keep the source licenses (non-commercial for the Loh et al. data).

Acknowledgements

We thank the organizers of the DBCLS BioHackathon 2026, and the authors of the datasets we used for making their data openly available.

References

- Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., & Grundmann, M. (2020). BlazePose: On-device real-time body pose tracking. <https://doi.org/10.48550/ARXIV.2006.10204> [cito:usesMethodIn].
- Fukuchi, R., Fukuchi, C., & Duarte, M. (2023). A public data set of running biomechanics and the effects of running speed on lower extremity kinematics and kinetics. <https://doi.org/10.6084/M9.FIGSHARE.4543435> [cito:citesAsDataSource].
- Fukuchi, R. K., Fukuchi, C. A., & Duarte, M. (2017). A public dataset of running biomechanics and the effects of running speed on lower extremity kinematics and kinetics. *PeerJ*, 5, e3298. <https://doi.org/10.7717/peerj.3298> [cito:citesAsDataSource].
- Harrington, M., Zavatsky, A., Lawson, S., Yuan, Z., & Theologis, T. (2007). Prediction of the hip joint centre in adults, children, and patients with cerebral palsy based on magnetic resonance imaging. *Journal of Biomechanics*, 40(3), 595–602. <https://doi.org/10.1016/j.jbiomech.2006.02.003> [cito:usesMethodIn].
- Loh, R. B. C., & Kong, P. W. (2026). Discriminant efficacy of video-based gait analysis in running-related injuries. *Journal of Medical and Biological Engineering*, 46(1), 100–108. <https://doi.org/10.1007/s40846-026-01008-y> [cito:citesAsDataSource].
- Loh, R. B. C., Pan, J. W., Iskandar, M. N. S., & Kong, P. W. (2025). Predicting running-related injuries from functional, kinetic and kinematic data. *International Journal of Sports Medicine*, 47(05), 372–380. <https://doi.org/10.1055/a-2706-5516> [cito:citesAsDataSource].
- Wang, H. (2022). RGB data [front and side views] of the wearable system validation study. <https://doi.org/10.5281/ZENODO.6644593> [cito:citesAsDataSource].
- Wang, H., Basu, A., Durandau, G., & Sartori, M. (2022). Comprehensive kinetic and EMG dataset of daily locomotion with 6 types of sensors. <https://doi.org/10.5281/ZENODO.6457662> [cito:citesAsDataSource].
- Wang, H., Basu, A., Durandau, G., & Sartori, M. (2023). A wearable real-time kinetic measurement sensor setup for human locomotion. *Wearable Technologies*, 4. <https://doi.org/10.1017/wtc.2023.7> [cito:citesAsRelated].
- Wu, H., Brooke-Wavell, K., Barnes, M. R., Awan, Z., Mastana, S., Allen, S., & Blagrove, R. C. (2026). Multidisciplinary prediction of running-related injuries using machine learning. *npj Digital Medicine*, 9(1). <https://doi.org/10.1038/s41746-026-02413-y> [cito:citesAsDataSource].
- Zeni, J., Richards, J., & Higginson, J. (2008). Two simple methods for determining gait events during treadmill and overground walking using kinematic data. *Gait & Posture*, 27(4), 710–714. <https://doi.org/10.1016/j.gaitpost.2007.07.007> [cito:usesMethodIn].